DIFFERENTIAL EQUATIONS LEC. NOTES WEEK IX

SOLVING NONHOMOGENEOUS LINEAR HIGHER ORDER ODEs

$$y^{(n)} + a_{n-1}(t)y^{(n-1)} + \cdots + a_1(t)y' + a_0(t)y = g(t)$$

$$L(y) = g(t).$$

$$y = y_h + y_p$$

y_h: solution for hom. D.E $L[y]=0$ (H)

We know how to solve with constant coefficient.

y_p: particular solution for $L[y]=g(t)$ (N) (Show with $Y(t)$)

For particular solution of $Y(t)$

we will use the following three methods

1. Undetermined coefficient technique
   (works only for some special $g(t)$ and for D.E with constant coefficients)

2. Variation of Parameters (VOP)
3. Reduction of order

$\}$ work for variable coefficient linear O.D.E and for any $g(t)$. However they require the solution of the corresponding homogene D.E $L[y]=0$ (H)

(I) UNDETERMINED COEFFICIENT TECHNIQUE (UCH)

is applicable when the coefficients in (N) are constants

and $g(t)$ is of a special form

That is (1) $a_i \in \mathbb{R}$ $\forall i = 0, 1, \ldots, n-1$

(2) $g(t) \in$ Class $= \{$ exp, sine, cosine, polynomials, and their sums, products$\}$

Ex $g(t) = \{ e^{-t}, \sin 3t, \cos 2t, t^2+1, t^2 e^{-3t}, e^{-t} \sin 2t \}$

HOW TO SOLVE WITH UCH

(1) Solve homogeneous equation with constant coefficients $L[y] = 0$ and find y_h (we know that step)

(2) If $g(t) = g_1(t) + g_2(t) + \cdots + g_n(t)$ where each $g_i(t) \in$ Class

then form n-subproblem. That is.

$$\underbrace{L[y] = g_1(t) \qquad L[y] = g_2(t) \qquad \cdots \qquad L[y] = g_n(t)}_{L[y] = g_i(t) \qquad i = 1, 2, \ldots n}$$

(3) Consider the ith problem $L[y] = g_i(t)$. If any dublication in the assumed form of particular solution $Y_i(t)$ with the solution of hom. eq. $L[y]=0$, then multiply $Y_i(t)$ by t or t^2 so as to avoid dublication

$g_i(t)$	λ	Y_i
$f(t) = b_p t^p + \dots + b_1 t + b_0$, $p \in \mathbb{N}$	$\lambda \neq 0$	$A_p t^p + A_{p-1} t^{p-1} + \dots + A_1 t + A$
$K.e^{at}$ $K, a \in \mathbb{R}$	$\lambda \neq a$	$A.e^{at}$
$P \sin bt + Q \cos bt$ $P, Q, b \in \mathbb{R}$	$\lambda \neq \mp ib$	$A \cos bt + B \sin bt$
$e^{at} f(t)$	$\lambda \neq a$	$e^{at}(A_p t^p + \dots + A_1 t + A_0)$
$e^{at}(P \sin bt + Q \cos bt)$	$\lambda \neq a \mp ib$	$e^{at}(A \cos bt + B \sin bt)$
$(P \sin bt + Q \cos bt) f(t)$	$\lambda \neq \mp ib$	$(A_p t^p + \dots + A_0) \cos bt + (B_p t^p + \dots + B_0) \sin bt$
$e^{at}(P \sin bt + Q \cos bt) f(t)$	$\lambda \neq a \mp ib$	$e^{at}[(A_p t^p + \dots + A_0) \cos bt + (B_p t^p + \dots + B_0) \sin bt]$

Ex $y'' - 2y' + y = e^t$

(1) y_h $\lambda^2 - 2\lambda + 1 = 0$ $(\lambda - 1)^2 = 0$ $\lambda_{1,2} = 1$.

$y_h = c_1 e^t + c_2 t e^t$

(2) $g(t) = e^t \rightarrow Y(t) = t.t\, A e^t$ (to avoid dublication)

$Y(t) = t^2 A e^t$

$y'' - 2y' + y = e^t$ $\quad y'(t) = 2t A e^t + t^2 A e^t$

$Y''(t) - 2Y'(t) + Y(t) = e^t$ $\quad y''(t) = 2A e^t + 2tA e^t + 2tA e^t + t^2 A e^t$

Then we get $e^t = 2A e^t$ $\quad A = 1/2$

$y(t) = 1/2\, t^2 e^t$ $\quad Y_{general} = y_h + Y(t) = c_1 e^t + c_2 t e^t + \frac{1}{2} t^2 e^t$

(2) $y'' - 3y' - 4y = \underbrace{3e^{2t}}_{g_1(t)} + \underbrace{2\sin t}_{g_2(t)} + \underbrace{8e^{t}\cos 2t}_{g_3(t)}.$

$\underline{y_h}$ $\lambda^2 - 3\lambda - 4 = 0$ $\quad (\lambda-4)(\lambda+1) = 0 \quad \lambda_1 = 4 \quad \lambda_2 = -1$

$F.S.S = \{e^{-t}, e^{4t}\}.$

$\underline{Y(t)}$ $\quad g_1(t) = 3e^{2t} \Rightarrow Y_1(t) = A.e^{2t}$

$g_2(t) = 2\sin t \Rightarrow Y_2(t) = B\sin t + C\cos t$

$g_3(t) = 8e^{t}\cos 2t \Rightarrow Y_3(t) = De^{t}\cos 2t + Ee^{t}\sin 2t.$

$Y_1''(t) - 3Y_1'(t) - 4Y_1(t) = 3e^{2t}.$

$4Ae^{2t} - 6Ae^{2t} - 4A.e^{2t} = 3e^{2t} \quad A = -1/2.$

$Y_1(t) = -1/2\, e^{2t}$

$\underbrace{Y_2''(t)}\, - 3Y_2'(t) - 4Y_2(t) = 2\sin t$

$(-B\sin t - C\cos t) - 3(B\cos t - C\sin t) - 4(B\sin t + C\cos t)$

$(-B + 3C - 4B)\sin t + (-C - 3B - 4C)\cos t = 2\sin t.$

$-5B + 3C = 2. \qquad A = -5/17 \quad B = 3/17$

$-3B - 5C = 0.$

$Y_2(t) = -\frac{5}{17}\sin t + \frac{3}{17}\cos t$

Exercise find $\quad Y_3'' - 3Y_3' - 4Y_3 = -8e^{t}\cos 2t.$

$D = \frac{10}{13} \quad E = 2/13 \quad Y_3(t) = \frac{10}{13}e^{t}\cos 2t + \frac{12}{13}e^{t}\sin 2t$

Then, $Y(t) = Y_1(t) + Y_2(t) + Y_3(t)$

$y_g = y_h + Y(t) = C_1e^{-t} + C_2e^{4t} - \frac{1}{2}e^{2t} - \frac{5}{17}\sin t$

$+ \frac{3}{17}\cos t + \frac{10}{13}e^{t}\cos 2t.$

$+ \frac{2}{13}e^{t}\sin 2t.$

Ex Find the form of particular solutions

1) $y''' + 2y'' - y' - 2y = e^t + t^2$.

$\underline{y_h}$ $\lambda^3 + 2\lambda^2 - \lambda - 2 = 0 \Rightarrow \lambda^2(\lambda+2) - (\lambda+2) = 0 \quad (\lambda^2-1)(\lambda+2) = 0$

$\lambda = \mp 1 \qquad \lambda = -2$

F.S.S $= \{e^{-2t}, e^{-t}, e^t\}$.

$\underline{Y(t)}$ $g(t) = e^t + t^2$

$t \cdot Ae^t \qquad (A_0 + A_1 t + A_2 t^2)$

$Y(t) = A \cdot t \cdot e^t + A_0 + A_1 t + A_2 t^2$.

2) $y^{(4)} + 2y'' + y = 3\sin t - 5\cos t$

$\underline{y_h}$ $\lambda^4 + 2\lambda^2 + 1 = 0$.

$\lambda^2(\lambda^2+1) + \lambda^2 + 1 = 0$.

$(\lambda^2+1)^2 = 0 \qquad \lambda = \mp i, \ \lambda = \mp i$

F.S.S $= \{e^{0t}\cos t, e^{0t}\sin t, t \cdot e^{0t}\cos t, t \cdot e^{0t}\sin t\}$.

$\underline{Y(t)}$ $g(t) = 3\sin t - 5\cos t$

$t^2(A\sin t + B\cos t)$

3) $y''' - 2y' + 5y = t^2 + 2t \cdot e^t \sin 2t$.

$\lambda^2 - 2\lambda + 5 = 0 \qquad \lambda_{1,2} = \dfrac{2 \mp \sqrt{4-20}}{2} = 1 \mp 2i$

$\underline{y_h}$ $\{e^t\cos 2t, e^t \sin 2t\}$.

$\underline{Y(t)}$ $t^2 + 2te^t \sin 2t$

$\longrightarrow [(A_0 + B_1 t)e^t \sin 2t + (B_0 + C_1 t)e^t \cos 2t]t$

$A_0 + A_1 t + A_2 t^2$.

$Y(t) = A_0 + A_1 t + A_2 t^2 + [(B_0 + B_1 t)e^t \sin 2t + (C_0 + C_1 t)e^t \cos 2t]t$

④ $y'' - 2y' - 3y = t.e^{-t} - e^{-t} + t^2 - 1$

y_h $\lambda^2 - 2\lambda - 3 = 0$

$(\lambda-3)(\lambda+1) = 0 \Rightarrow \lambda = 3, \lambda = -1$ $F.S.S = \{e^{-t}, e^{3t}\}$

$y(t) \Rightarrow g(t) = \underbrace{t.e^{-t} - e^{-t}}_{(t-1)e^{-t}} + \underbrace{t^2 - 1}_{B_0 + B_1 t + B_2 t^2}$

$\Downarrow$

$t.(A_0 + A_1 t)e^{-t}$

$y(t) = t(A_1 t + A_0)e^{-t} + B_2 t^2 + B_1 t + B_0$

⑤ $y^{(4)} - y = t.e^{t} + t\cos t$

y_h $\lambda^4 - 1 = 0$ $F.S.S = \{e^{t}, e^{-t}, \cos t, \sin t\}$

$(\lambda^2-1)(\lambda^2+1) = 0$

$\lambda = \mp 1$ $\lambda = \mp i$

y $e^{t}t$: $y_1(t) = t(A_0 + A_1 t)e^{t}$

$t\cos t$: $y_2(t) = t(B_0 + B_1 t)\cos t + t(C_0 + C_1 t)\sin t$

⑥ $y'' + 2y' + 2y = x + 1$

y_h $\lambda^2 + 2\lambda + 2 = 0$ $\lambda_{1,2} = \frac{-2 \mp \sqrt{4 - 4.2}}{2} = -1 \mp i$

$F.S.S = \{e^{-x}\cos x, e^{-x}\sin x\}$

$y(t)$ $y(t) = A_0 + A_1 x$

⑦ $y''' - 3y'' + 3y' - y = 2e^{x}$

y_h $\lambda^3 - 3\lambda^2 + 3\lambda - 1 = 0$ $F.S.S = \{e^{x}, xe^{x}, x^2 e^{x}\}$

$(\lambda - 1)^3 = 0$

$y(t)$ $y(t) = x^3.A.e^{x}$

(2) VARIATION OF PARAMETERS (VOP)

Applicable for linear homogeneous DE with variable coefficients. But we need to know the solution for homogeneous D.E. VOP is applicable for any $g(t)$

(H) $L[y]=0$

(N) $L[y]=g(t)$

$y_h = c_1y_1+c_2y_2+\dots+c_ny_n$ where y_p are solutions of (H)

Then, $Y(t) = u_1(t)\,y_1(t) + u_2(t)y_2(t)+\dots+u_n(t)\,y_n(t)$ (1)

Then, substitute (1) to (N) to obtain the equations of VOP technique

Firstly, we will discuss VOP for 2^{nd} order ODEs.

(*) $y'' + p(t)y' + q(t)y = g(t)$ "2nd order linear ODE" non-hom.

Suppose that $y_h = c_1y_1(t)+c_2y_2(t)$ such that $y_1(t)$ and $y_2(t)$ are solutions of (H)

Then, we assume $Y(t) = u_1(t)y_1(t)+u_2(t)y_2(t)$. Substitute in (*)

$Y'(t) = \underbrace{\boxed{u_1'y_1+u_2'y_2}}_{\text{Force to be } 0} + u_1y_1'+u_2y_2'$ $\Rightarrow$ $\boxed{u_1'y_1+u_2'y_2=0}$ (1)

$Y''(t) = u_1'y_1'+u_2'y_2'+u_1y_1''+u_2y_2''$

Subs in (*): $Y''(t) + P(t)Y'(t) + q(t)Y(t) = g(t)$

$u_1'y_1'+u_2'y_2'+u_1y_1''+u_2y_2''+p(t)[u_1y_1'+u_2y_2']+q(t)[u_1y_1+u_2y_2]=g(t)$

$u_1\underbrace{[y_1''+p(t)y_1'+q(t)y_1]}_{\substack{=0 \\ \text{since } L[y_1]=0,\\ y_1 \text{ is solution of hom D.E}}}+u_2\underbrace{[y_2''+p(t)y_2'+q(t)y_2]}_{\substack{=0 \text{ since } y_2 \text{ is a sol.}\\ \text{of (H)}}}+u_1'y_1'+u_2'y_2'=g(t)$

$\boxed{u_1'y_1'+u_2'y_2'=g(t)}$ (2)

Then, the eqs. of VoP.

① $u_1' y_1 + u_2' y_2 = 0$

② $u_1' y_1' + u_2' y_2' = p(t)$

$$\Rightarrow \underbrace{\begin{bmatrix} y_1 & y_2 \\ y_1' & y_2' \end{bmatrix}}_{\psi(t)} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ p(t) \end{bmatrix}$$

Note that $\psi^{-1} = \dfrac{1}{\det \psi} \begin{bmatrix} y_2' & -y_2 \\ -y_1' & y_1 \end{bmatrix}$

$\det(\psi) = W(y_1, y_2) \neq 0$ since y_1, y_2 are linearly independent solutions of (H)

Then, the system has a unique solution

$$\begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \frac{1}{W(y_1, y_2)} \begin{bmatrix} y_2' & -y_2 \\ -y_1' & y_1 \end{bmatrix} \begin{bmatrix} 0 \\ p(t) \end{bmatrix} = \frac{1}{W} \begin{bmatrix} -y_2 g(t) \\ y_1 g(t) \end{bmatrix}$$

$$u_1' = -\frac{y_2 g(t)}{W} \Rightarrow u_1(t) = \int -\frac{y_2 g(t)}{W}\, dt.$$

$$u_2' = \frac{y_1 p(t)}{W} \Rightarrow u_2(t) = \int -\frac{y_1 p(t)}{W}\, dt.$$

$$Y(t) = u_1 y_1 + u_2 y_2 = y_1 \int -\frac{y_2 g(t)}{W}\, dt + y_2 \int \frac{y_1 p(t)}{W}\, dt$$

Note that This formula will give us the general solution if we keep the integration constants different than zero.

On the other hand if we set integration constants to zero, we end up with a particular solution $Y(t)$ and

$$y_g(t) = Y(t) + y_h$$

Summary Equations of VOP $\boxed{\begin{aligned} u_1' y_1 + u_2' y_2 &= 0 \\ u_1' y_1' + u_2' y_2' &= p(t) \end{aligned}}$

Ex: $y'' + 2y' + y = \frac{e^{-t}}{t}$ Solve by VOP

$\underline{y_h}$ $\lambda^2 + 2\lambda + 1 = 0 \Rightarrow (\lambda+1)^2 = 0$ $\lambda = -1$

$y_h = c_1 e^{-t} + c_2 t \cdot e^{-t}$

$\underline{Y(t)}$ By VOP $Y(t) = u_1 e^{-t} + u_2 t e^{-t}$

$\underline{\text{VOP eqs}}$ $u_1' e^{-t} + u_2' e^{-t} \cdot t = 0$

$u_1'(-e^{-t}) + u_2'(e^{-t} - te^{-t}) = \frac{e^{-t}}{t}$

$$\begin{bmatrix} e^{-t} & t \cdot e^{-t} \\ -e^{-t} & e^{-t} - te^{-t} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \end{bmatrix} = \begin{bmatrix} 0 \\ e^{-t}/t \end{bmatrix}$$

Let's use Cramer's Rule

$u_j' = \frac{|\Psi_j|}{W}$ Ψ_j is the matrix obtained by changing jth column of Ψ with rhs vector

$W = \det \Psi$

$$u_1' = \frac{\begin{vmatrix} 0 & te^{-t} \\ e^{-t}/t & e^{-t} - te^{-t} \end{vmatrix}}{\begin{vmatrix} e^{-t} & te^{-t} \\ -e^{-t} & e^{-t} - te^{-t} \end{vmatrix}} = \frac{-e^{-2t}}{e^{-2t}} = -1 \Rightarrow u_1' = -1 \quad u_1(t) = -t + c_1$$

$$u_2' = \frac{\begin{vmatrix} e^{-t} & 0 \\ -e^{-t} & e^{-t}/t \end{vmatrix}}{e^{-2t}} = \frac{e^{-2t}/t}{e^{-2t}} = 1/t \quad u_2 = \ln|t| + c_2$$

Then, $Y_{gen}(t) = u_1 y_1 + u_2 y_2 = (t + c_1) y_1 + (\ln|t| + c_2) y_2$

$$= \underbrace{t y_1 + \ln|t| y_2}_{Y(t) \text{ specific particular solution}} + \underbrace{c_1 y_1 + c_2 y_2}_{y_h}$$

Ex $y''+y=\tan t$ Solve with VOP.

y_h $\lambda^2+1=0$ $\lambda=\mp i$ $y_h=c_1\cos t+c_2\sin t$

$Y(t)$ $Y(t)=u_1\cos t+u_2\sin t.$

VOP. Eq. $u_1'\cos t+u_2'\sin t=0$

$u_1'(-\sin t)+u_2'(\cos t)=\tan t$ $\Rightarrow \begin{bmatrix}\cos t & \sin t\\ -\sin t & \cos t\end{bmatrix}\begin{bmatrix}u_1'\\u_2'\end{bmatrix}=\begin{bmatrix}0\\ \tan t\end{bmatrix}$

$$u_1'=\frac{\begin{vmatrix}0 & \sin t\\ \tan t & \cos t\end{vmatrix}}{1}=\tan t\sin t=\frac{\sin^2 t}{\cos t}\Rightarrow u_1(t)=\int\frac{\overbrace{\sin^2 t}^{1-\cos^2 t}}{\cos t}dt.$$

$$u_1(t)=\int\left(\frac{1}{\cos t}-\cos t\right)dt=\int(\sec t-\cos t)dt$$
$$=\ln|\sec t+\tan t|-\sin t+\underset{0}{\underset{\shortparallel}{c_1}}$$

$$u_2'(t)=\frac{\begin{vmatrix}\cos t & 0\\ -\sin t & \tan t\end{vmatrix}}{1}=\cos t\tan t=\sin t \qquad u_2(t)=-\cos t+\underset{0}{\underset{\shortparallel}{c_2}}$$

$$Y(t)=(\ln|\sec t+\tan t|-\sin t)\underbrace{\cos t}_{y_1}+(-\cos t)\underbrace{\sin t}_{y_2}$$

So $y_{general}=Y(t)+c_1\cos t+c_2\sin t.$

2 VOP WITH HIGHER ORDER ODEs

$$y^{(n)}+a_{n-1}(t)y^{(n-1)}+\cdots+a_1(t)y'+a_0(t)y=p(t)$$

$$y_h=c_1y_1+c_2y_2+\cdots+c_ny_n$$
$$Y(t)=u_1(t)y_1+u_2(t)y_2+\cdots+u_n(t)y_n$$

Eqns of VOP
$$u_1'y_1+u_2'y_2+\cdots+u_n'y_n=0$$
$$u_1'y_1'+u_2'y_2'+\cdots+u_n'y_n'=0$$
$$u_1'y_1''+u_2'y_2''+\cdots+u_n'y_n''=0$$
$$\vdots$$
$$u_1'y_1^{(n-1)}+u_2'y_2^{(n-1)}+\cdots+u_n'y_n^{(n-1)}=p(t)$$

$$\underbrace{\begin{bmatrix} y_1 & y_2 & \cdots & y_n \\ y_1' & y_2' & \cdots & y_n' \\ \vdots & & & \\ y_1^{(n-1)} & y_2^{(n-1)} & \cdots & y_n^{(n-1)} \end{bmatrix}}_{\psi} \begin{bmatrix} u_1' \\ u_2' \\ \vdots \\ u_n' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ p(t) \end{bmatrix}$$

$$u_j' = \frac{|\psi_j|}{W(y_1, y_2, \ldots, y_n)}$$

Ex Solve $y''' - y'' - y' + y = e^{2t}$ by VOP

<u>y_h</u> $\lambda^3 - \lambda^2 - \lambda + 1 = 0$ $\quad (\lambda-1)(\lambda^2-1) = 0 \Rightarrow \lambda = 1$ $alg(1) = 2$, $\lambda = -1$

$y_h = c_1 e^t + c_2 t e^t + c_3 e^{-t}$

$y(t) = u_1 e^t + u_2 t e^t + u_3 e^{-t}$

<u>VOP Equations</u>

$$\begin{bmatrix} e^t & te^t & e^{-t} \\ e^t & (e^t + te^t) & -e^{-t} \\ e^t & 2e^t + te^t & e^{-t} \end{bmatrix} \begin{bmatrix} u_1' \\ u_2' \\ u_3' \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ e^{2t} \end{bmatrix}$$

$$u_1'(t) = \frac{\begin{vmatrix} 0 & te^t & e^{-t} \\ 0 & te^t + e^t & -e^{-t} \\ e^{2t} & 2e^t + te^t & e^{-t} \end{vmatrix}}{W(te^t, e^t, e^{-t})} = \frac{1}{4} e^t(-2t-1) \Rightarrow u_1(t) = -\frac{1}{2} te^t + \frac{1}{4} e^{-t} + \underset{0}{\overset{\shortparallel}{c_1}}$$

$$u_2'(t) = \frac{\begin{vmatrix} e^t & 0 & e^{-t} \\ e^t & 0 & -e^{-t} \\ e^t & e^{2t} & e^{-t} \end{vmatrix}}{4e^t} = \frac{1}{2} e^t \quad u_2(t) = \frac{1}{2} e^t + \underset{0}{\overset{\shortparallel}{c_2}}$$

$u_3' = \frac{1}{4} e^{3t} \rightarrow u_3 = \frac{1}{2} e^{3t} + \underset{0}{\overset{\shortparallel}{c_3}}$

$$y(t) = \left(\frac{1}{2} te^t + \frac{1}{4} e^{-t}\right)\overset{y_1}{\overset{\shortparallel}{e^t}} + \frac{1}{2} e^t \underbrace{te^t}_{y_2} + \frac{1}{12} e^{3t} \underbrace{e^{-t}}_{y_3}$$

$y_p = y(t) + y_h.$

3) REDUCTION OF ORDER

Requires knowing one solution of the corresponding homogeneous D.E (2nd order D.E)

$$y'' + p(t)y' + q(t)y = 0 \quad (H)$$

$$y'' + p(t)y' + q(t)y = g(t) \quad (N)$$

Let y_1 be a solution of (H)

Set $y(t) = u(t).y_1(t) \xrightarrow[\text{subs in } (N)]{} y'' + p(t)y' + q(t)y = g(t)$

$$y' = u'y_1 + u\,y_1'$$

$$y'' = u''y_1 + 2u'y_1' + u\,y_1''$$

$$y'' + p(t)y' + q(t)y = g(t)$$

$$u''y_1 + 2u'y_1' + u\,y_1'' + p(t)(u'y_1 + uy_1') + q(t)uy_1 = g(t)$$

$$u\underbrace{(y_1'' + p(t)y_1' + q(t)y_1)}_{0 \text{ since } y_1 \text{ is solution of } (H)} + y_1u'' + (2y_1' + p(t)y_1)u' = g(t)$$

$$y_1u'' + (2y_1' + p(t)y_1)u' = g(t)$$

Use subs. $v = u' \Rightarrow v' = u''$

$$y_1v' + (2y_1' + p(t)y_1)v = g(t)$$

$$v' + \frac{(2y_1' + p(t)y_1)}{y_1}v = \frac{g(t)}{y_1}$$

$$\mu = e^{\int \frac{2y_1' + p(t)y_1}{y_1}dt}$$

We can obtain v, then $u' = v \Rightarrow u = \int v\,dt$. $y(t) = uy_1$

Ex $y'' - \frac{2}{t^2} y = 1$ Solve by using reduction of order if $y_1 = t^2$ is a solution of H.

$y(t) = u t^2 = t^2 u$

$y' = 2t u + t^2 u'$

$y'' = 2u + 4tu' + t^2 u''$

$y'' - \frac{2}{t^2} y = 1$

$\Downarrow$

$2u + 4tu' + t^2 u'' - \frac{2}{\cancel{t^2}}(u \cancel{t^2}) = 1$

$t^2 u'' + 4tu' = 1$

$u'' + \frac{4}{t} u' = \frac{1}{t^2}$ Let $v = u'$, $v' = u''$

$v' + \frac{4}{t} v = \frac{1}{t^2}$ "First order linear"

$\mu(t) = e^{\int 4/t \, dt} = e^{4 \ln t} = t^4$

$\underbrace{t^4 v' + 4t^3 v}$ $= t^2$

$\int (t^4 v)' \, dt = \int t^2 dt.$

$t^4 . v = \frac{t^3}{3} + C_1$

$v = \frac{1}{3t} + C_1 . t^{-4}$

$u' = v = \frac{1}{3t} + C_1 . t^{-4}$

$u(t) = \frac{1}{3} \ln|t| + C_1 . \frac{t^{-3}}{-3} + C_2$

$\Rightarrow y(t) = t^2 u$

$y(t) = \underbrace{\frac{t^2}{3} \ln|t|}_{\bar{y}(t)} + \underbrace{C_1 . \frac{t^{-1}}{-3}}_{y_2 \text{ sol of (H)}} + C_2 \underbrace{t^2}_{y_1}$

Since we keep integration constant we end up with general solution.

Ex 7 Solve $y''+2y'+y=\frac{e^{-t}}{t}$ by Reduction of order.

$p(t)=2 \quad q(t)=1. \quad g(t)=e^{-t}/t$

get y_1 $\quad r^2+2r+1=0 \quad (r+1)^2=0 \quad$ F.S.S $=\{e^{-t}, te^{-t}\}$

$r=-1,-1$

$y_1=e^{-t}$

Set $\quad Y(t)=u(t)\,y_1(t)=e^{-t}u(t) \qquad y_1'=-e^{-t}$

$v=u'$

$$v'+\frac{2y_1'+p(t)y_1}{y_1}v=\frac{g(t)}{y_1}$$

$$v'+\frac{(-2e^{-t}+2.e^{-t})}{e^{-t}}=\frac{\cancel{e^{-t}}}{t.\cancel{e^{-t}}}$$

$v'=\frac{1}{t} \qquad v(t)=\ln|t|+C_1$

$u'(t)=\ln|t|+C_1$

$u(t)=\ln|t|.t-t+C_1t+C_2.$

$$\int \ln t.dt=\ln t.t-\int\frac{1}{t}.t\,dt=\ln t.t-t$$

$\ln t=u \qquad dt=dv$

$\frac{1}{t}dt=du \qquad t=v$

$$Y(t)=Y_1(t).u(t)=e^{-t}(\ln|t|.t-t+C_1t+C_2)$$

$$=\underbrace{e^{-t}.t.\ln|t|}_{\tilde{y}}-e^{-t}.t+C_1\underbrace{t.e^{-t}}_{\text{other Solution of (H) } y_2(t)}+C_2\underbrace{e^{-t}}_{\text{solution of (H) } y_1(t)}$$